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PDANA8411w Part 1 Report

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Introduction

Insurance charge prediction is a vital analysis that helps health care providers to set their premiums for the best price and help clients understand what will contribute to the amount they pay. Statistical learning techniques as well as linear regression are very common to use to model relationships between an outcome and one or more predictor(s) within a dataset where there are both continuous and categorical data types. The linear relationship is the most basic method used to predict a dependent variable from an independent variable through the sum of the products of each independent feature (Rao et al., 2023; Taloba et al., 2022; Tiwary, 2021). This project analyzed, examined, and developed a predictive model that uses demographic/lifestyle variables to explain insurance charges using the Kaggle Medical Insurance dataset.

This report documents the steps taken to prepare the data for model development, to explore/analyze the data, to train models, to evaluate models, and finally to interpret results. The discussion of findings will be written in such a way so that it will be understandable by a business client who has no technical knowledge. The evidence from the statistical modeling literature support all the practices performed, including data transformations, feature selection, and how to interpret a regression equation (Cenita, Asuncion & Victoriano, 2023; Dwikasaria, 2023; Tiwary, 2021).

Dataset Overview and Suitability

The dataset analyzed for this project has 1,338 rows of data and includes seven columns with data regarding the patient's demographics (age, gender, weight/BMI) along with their family size (number of children), smoking habits, region, and charges. Because "charges" are a continuous measure (the target variable) it was determined that this dataset is ideal for the use of regression analysis (Cheverne, 2017). Continuous measures such as age and BMI may be entered directly into a regression model whereas categorical measures (such as gender, whether or not the individual smokes, and the region) will require some type of encoding to convert them to a format that can be modeled by the computer (Cenita, Asuncion & Victoriano, 2023; Cheverne, 2017; Tiwary, 2021).

It should also be noted that there were no missing data entries within the dataset. This eliminated the need for imputation and made the pre-processing portion of the research process significantly easier than if missing data existed.

As discussed previously (Tiwary, 2021; Dwikasaria, 2023; Cenita, Asuncion & Victoriano, 2023), one of the most important aspects of performing statistical modeling is first understanding how your data is structured and what types of data exist. In addition to being

aware of the data structure and variable types prior to applying regression models, you must ensure that both linear relationships exist between your independent and dependent variables and that the independent variable(s) are numeric.

Insurance claims and therefore healthcare costs typically have an uneven distribution across all individuals and groups, resulting in skewed distributions and outliers within insurance claim amounts. Although the existence of these phenomena does not eliminate the use of linear regression methods when examining healthcare cost data properly examined and interpreted they may provide useful insights (Taloba et al., 2022; Rao et al., 2023).

This report is intended to detail the steps taken during to complete this project including preparing data for analysis (preparation), analyzing data to determine patterns/relationships (exploration), developing, testing, evaluating and interpreting a predictive model. It is also expected that this documentation will be written in terms that are accessible to a client who has little technical experience. Research studies using statistical modeling techniques support many of the procedures used in this study including transforming data, selecting relevant data to include in analyses, and determining how to interpret results from regression analysis (Dwikasaria, 2023; Cenita, Asuncion & Victoriano, 2023; Tiwary, 2021).

Figure 1 – Dataset Summary

	age	sex	bmi	children	smoker	region	charges
1312	33	female	26.695	0	no	northwest	4571.41305
1313	34	male	42.9	1	no	southwest	4536.259
1314	19	female	34.7	2	yes	southwest	36397.576
1315	30	female	23.655	3	yes	northwest	18765.87545
1316	18	male	28.31	1	no	northeast	11272.33139
1317	19	female	20.6	0	no	southwest	1731.677
1318	18	male	53.13	0	no	southeast	1163.4627
1319	35	male	39.71	4	no	northeast	19496.71917
1320	39	female	26.315	2	no	northwest	7201.70085
1321	31	male	31.065	3	no	northwest	5425.02335
1322	62	male	26.695	0	yes	northeast	28101.33305
1323	62	male	38.83	0	no	southeast	12981.3457
1324	42	female	40.37	2	yes	southeast	43896.3763
1325	31	male	25.935	1	no	northwest	4239.89265
1326	61	male	33.535	0	no	northeast	13143.33665
1327	42	female	32.87	0	no	northeast	7050.0213
1328	51	male	30.03	1	no	southeast	9377.9047
1329	23	female	24.225	2	no	northeast	22395.74424
1330	52	male	38.6	2	no	southwest	10325.206
1331	57	female	25.74	2	no	southeast	12629.1656
1332	23	female	33.4	0	no	southwest	10795.93733
1333	52	female	44.7	3	no	southwest	11411.685
1334	50	male	30.97	3	no	northwest	10600.5483
1335	18	female	31.92	0	no	northeast	2205.9808
1336	18	female	36.85	0	no	southeast	1629.8335
1337	21	female	25.8	0	no	southwest	2007.945
1338	61	female	29.07	0	yes	northwest	29141.3603

Exploratory Data Analysis (EDA)

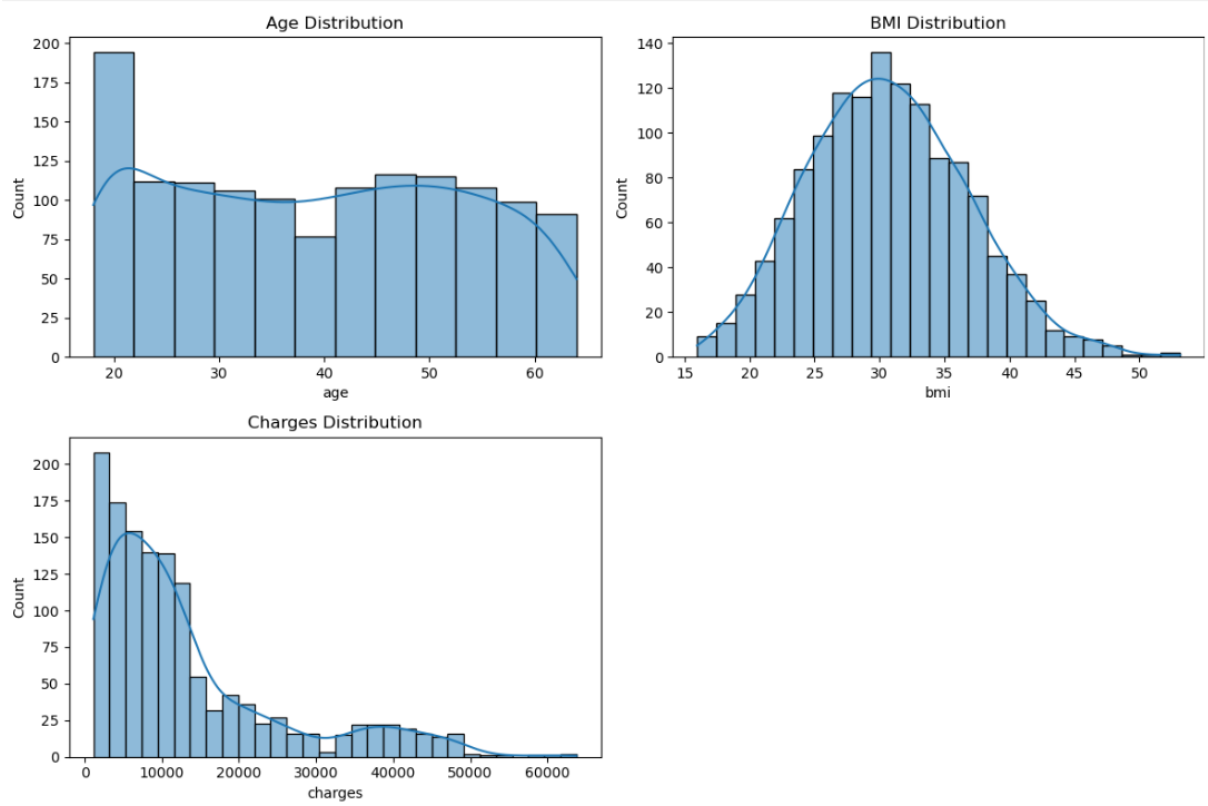
Descriptive Analysis and Distribution

The purpose of the descriptive statistical analysis was to describe the type of information contained within the dataset. The results from the summary statistics indicated that "charges" had considerable variability and therefore a wide variation of cost for each individual. To further assess the types of distributions, histograms and box plots were utilized:

- Age is generally uniformly distributed.
- BMI appears to be slightly right skewed due to the presence of extreme high values.
- Charges exhibit strong positive skewness which is typical in cost related data (Taloba et al., 2022; Rao et al., 2023).

Extreme values or outliers found in both BMI and Charges were recorded. These will need to be evaluated since their presence could potentially impact the performance of the trained model. An understanding of the distribution of variables is critical for ensuring that all necessary modeling assumptions have been met. For example, it is important to establish whether linear relationships exist as well as whether the residuals of the model meet the assumption of homoscedasticity (Rao et al., 2023; Taloba et al., 2022).

Figure 2 – Histograms



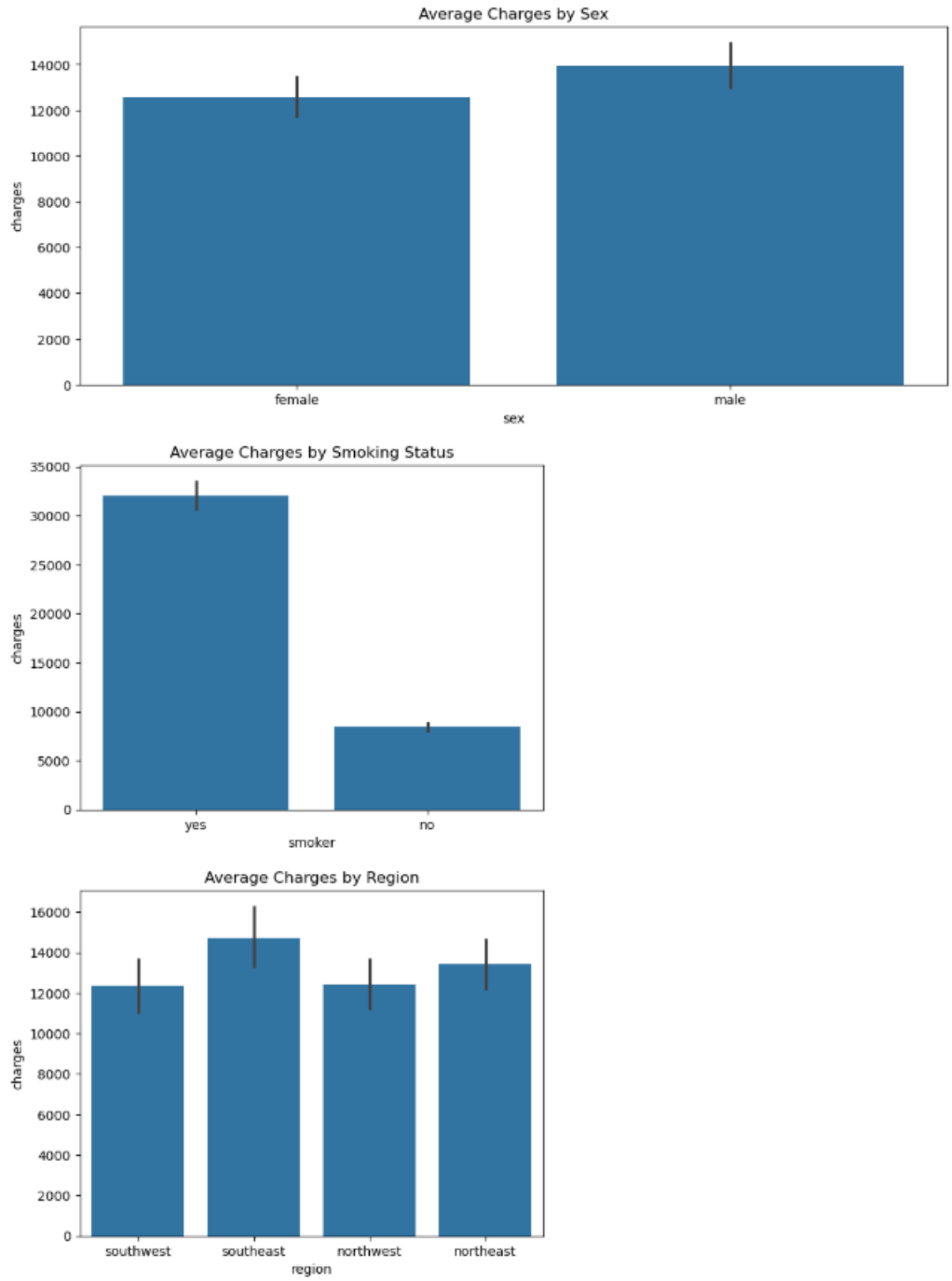
Categorical Variable Patterns

The categorical variables in this study were evaluated through comparative analyses across categories:

- Non-smoking was associated with lower mean insurance claims than smoking status, which is consistent with literature regarding increased health care expenditures as a result of smoking status (Cenita, Asuncion & Victoriano, 2023; Dwikasaria, 2023; Rao et al., 2023).
- Sex and region revealed less variation in the mean insurance claims.

These results support the reasoning to retain some of the factors within the model and to assess the degree of influence on outcomes from those that have been identified as being of lesser strength.

Figure 3 – Bar Charts of Sex, Smoking status and Region

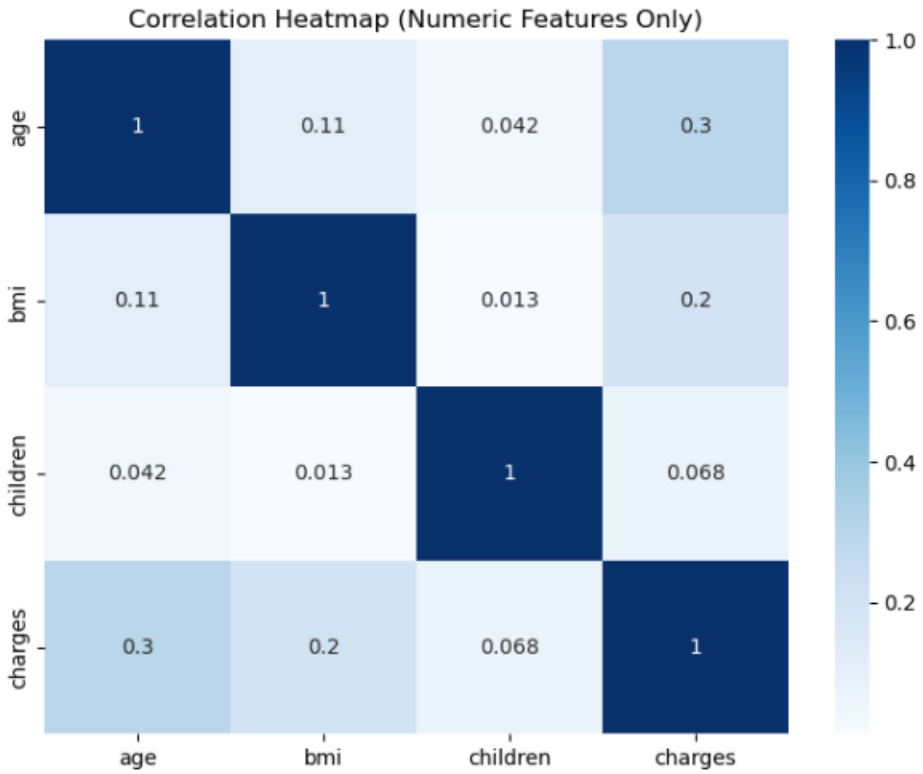


Correlation Analysis

There was a statistically significant correlation for the relationship of the predictor variables and the target variable in this study using numerical data:

- Correlations for BMI and age were moderate and positively correlated with charges.
- The number of children had weak correlations.

These results are consistent with other studies that have applied regression models to insurance-related databases (Dwikasaria, 2023; Cheverne, 2017; Cenita, Asuncion & Victoriano, 2023).



Feature Selection

Categorical features were converted into "one-hot" format to prevent categorical values from entering the regression model as non-numeric input. In addition, the selection of features was primarily based on both prior knowledge regarding how the field has been modeled, along with empirical observation of correlations. Both sex and region were included in the basic model because each could potentially affect charge, but would be evaluated for possible elimination or modification since neither sex nor region demonstrated a comparable level of influence upon charge data compared to smoking status (Cheverne, 2017; Dwikasaria, 2023; Cenita, Asuncion & Victoriano, 2023).

Removing the first one hot encoding was used for removing the redundancy of creating duplicate dummy variables via the use of one hot encoding to avoid multicollinearity. One hot encoding is a common methodology employed when modeling categorical variables utilizing regression models (Tiwary, 2021; Cenita, Asuncion & Victoriano, 2023).

Model Training and Implementation

A linear regression model was created by using a 80% / 20% train test split, in line with previously described methods for ensuring that models are unbiased (Cheverne, 2017; Cenita, Asunción & Victoriano, 2023), when creating and testing a model. Due to the fact that this type of model is appropriate for predicting a continuous outcome variable and has easily interpretable coefficients, we chose to use the LinearRegression class from the scikit-learn library.

The relationship between each independent variable and the charge as defined by the coefficient values will be evaluated in order to support and provide evidence-based interpretive insight into the features effect on charges, rather than solely evaluating the predictive accuracy of the model. For instance, positive coefficients for smoking status and BMI indicate that both of these characteristics have been associated with higher healthcare costs based upon previous studies (Dwikasaria, 2023; Cenita, Asunción & Victoriano, 2023).

Model Evaluation and Interpretation

The model has been assessed with the following:

- **R^2 Score** – shows what portion of charge variation is due to predictor variation.
- **Mean Absolute Error (MAE)** – shows the average amount of absolute difference between true and estimated value.
- **Mean Squared Error (MSE)** – penalizes greater differences from the true value.
- **Root Mean Squared Error (RMSE)** – an easily interpreted measure of total error size.

All four are common measures for evaluating regression models and are often referenced in predictive health care cost studies.

Model Refinement

To find other means by which models could be made to perform better, a more detailed version of the model was built after eliminating some of the lower contributing statistical features like particular regional indicators. The comparative metrics demonstrated that the development of this revised model moderately increased certain comparison metric results, thereby showing the advantages of using a model with fewer variables.

Conclusions and Business Implications

The linear regression model provided a statistically sound method of developing a predictive model for insurance charges. Additionally, it was able to provide strong associations of various factors including BMI, smoking status, etc. These are examples of the risk factors that insurance companies may use when setting their premiums/policies, or as health interventions for customers (Cenita, Asunción & Victoriano, 2023; Rao et al., 2023; Taloba et al., 2022).

The model will allow stakeholders to have decision-making information in terms of premium pricing and/or risk assessments. It could be improved by comparing non-linear regression models, and/or adding additional features.

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